

DESIGN AND DEVELOPMENT OF AN IoT-ENABLED BARANGAY RESIDENT & ATTENDANCE MANAGEMENT SYSTEM

FIELD OF THE STUDY

This study focuses on the design and development of an IoT-enabled system aimed at improving resident tracking and attendance management within barangay operations. It involves the integration of biometrics, microcontroller nodes, local servers, and a comprehensive web-based application. The system combines both hardware and software components, including fingerprint scanners, an embedded microcomputer, real-time clock modules, and a browser-accessible interface, to provide a centralized and secure solution for barangay administration.

BACKGROUND OF THE STUDY

Resident record management and attendance tracking play a crucial role in maintaining order and efficiency within local government units, particularly at the barangay level. Existing systems often rely on manual logbooks or outdated paper-based filing, which are prone to human error, data loss, and inefficient retrieval. Manual attendance tracking requires personnel to physically verify identities and record time logs, which may lead to delays, inaccuracies, or unauthorized entries.

On the other hand, traditional digital systems often lack real-time synchronization, biometric security, and remote accessibility, making it difficult for barangay officials to monitor activities efficiently. Many existing solutions depend heavily on constant internet connectivity or lack offline timekeeping components, leading to data inconsistencies during power interruptions or network failures. Additionally, fragmented systems do not provide user-friendly interfaces that unify both resident profiling and attendance monitoring.

Given these limitations, there is a need for a more advanced, secure, and integrated solution that utilizes biometric authentication, accurate timekeeping, and accessible web dashboards. This study aims to address these challenges by

developing a system that enhances data precision, secures resident records, reduces manual intervention, and improves overall barangay operational efficiency.

SUMMARY OF THE STUDY

This study presents the design and development of an IoT-enabled barangay resident and attendance management system that provides accurate, secure, and efficient tracking of individuals in the community. The system utilizes an Orange Pi Zero microcomputer as the central local server, integrated with an ESP32S microcontroller and R307 fingerprint sensor for biometric authentication. A DS3231 real-time clock (RTC) module ensures precise offline timekeeping, while an I2C LCD 1602 display provides immediate feedback during attendance logging.

A comprehensive web-based interface allows barangay officials to manage resident profiles, review attendance logs, and configure system settings through a local network using a standard web browser. This enables easy access and real-time monitoring without requiring specialized software installation. The local database securely stores biometric templates and resident information, ensuring data privacy and fast retrieval.

Overall, the developed system offers a modern, user-friendly, and highly secure solution that improves attendance management, minimizes identity fraud, and enhances the administrative efficiency of barangay operations.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings illustrate the present invention. In such drawings:

Fig. 1 is a system overview block diagram of the complete IoT-enabled resident and attendance management system showing all major component interconnections, signal flow, and communication paths among the Orange Pi Zero server, ESP32S, R307 Fingerprint sensor, LCD display, RTC module, and web browser interface.

Fig. 2 is a detailed view of the Orange Pi Zero single-board computer showing its network ports, USB interfaces, and expansion headers.

Fig. 3 is a view of the DS3231 Real-Time Clock module showing the DS3231 IC chip, EEPROM, backup battery, and I2C pin header connections.

Fig. 4 is a view of the I2C LCD 1602 Display module showing the 16x2 character screen and the I2C interface backpack.

Fig. 5 is a view of the ESP32S microcontroller showing its Wi-Fi/Bluetooth antenna, GPIO pins, and processing module.

Fig. 6 is a view of the R307 Fingerprint optical sensor showing the scanning window, housing, and communication interface.

Fig. 7 is a view of the web-based user interface dashboard accessible through a browser, showing the resident statistics, attendance logs, and system navigation menu.

DETAILED DESCRIPTION OF THE INVENTION

Figure 1 illustrates the complete system overview of the IoT-Enabled Barangay Resident & Attendance Management System, showing all hardware components, their interconnections, and signal flow paths. The Orange Pi Zero serves as the central local server, hosting the main database and the web application. It communicates over a local area network with the ESP32S microcontroller, which acts as the biometric attendance node. The ESP32S interfaces with the R307 Fingerprint sensor to capture and verify resident biometrics, and it connects to the I2C LCD 1602 Display to provide immediate visual feedback to the user upon scanning. The DS3231 Real-Time Clock module is connected to the ESP32S via the I2C bus to ensure accurate timestamping of attendance events even when offline. A web browser running on any authorized device connected to the local network communicates with the Orange Pi Zero, allowing barangay officials to register new residents, manage biometric data, view attendance reports, and configure system settings seamlessly.

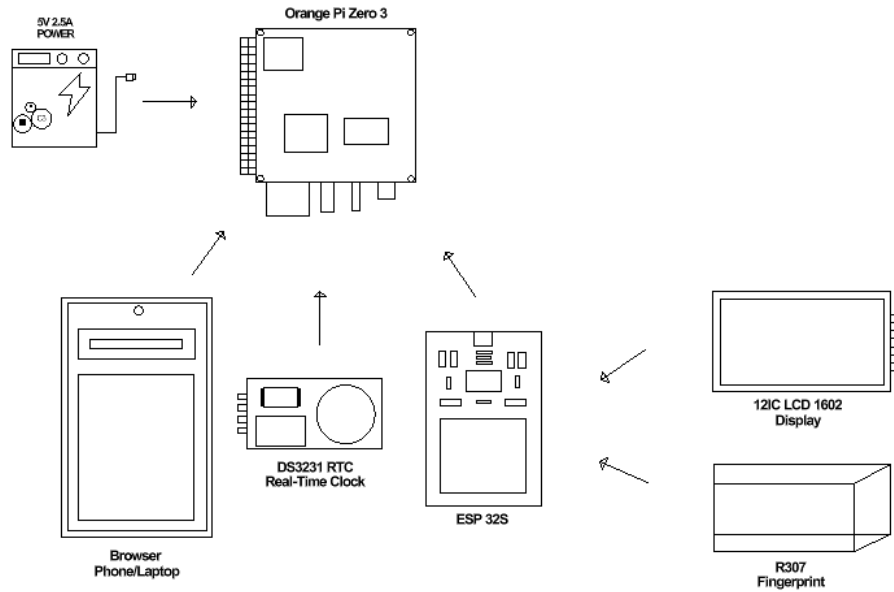


Fig. 1 System Overview

Figure 2 shows the Orange Pi Zero single-board computer, which serves as the main local server and processing unit of the system. The board features a compact form factor but provides sufficient processing power with its quad-core ARM Cortex-A7 processor. It hosts the centralized database where all resident records and attendance logs are securely stored. The built-in Ethernet port ensures stable and high-speed wired local area network connectivity, which is essential for serving the web application to multiple connected client devices simultaneously. The USB port allows for additional storage or peripheral connections, while the expansion headers provide flexibility for future hardware integrations.

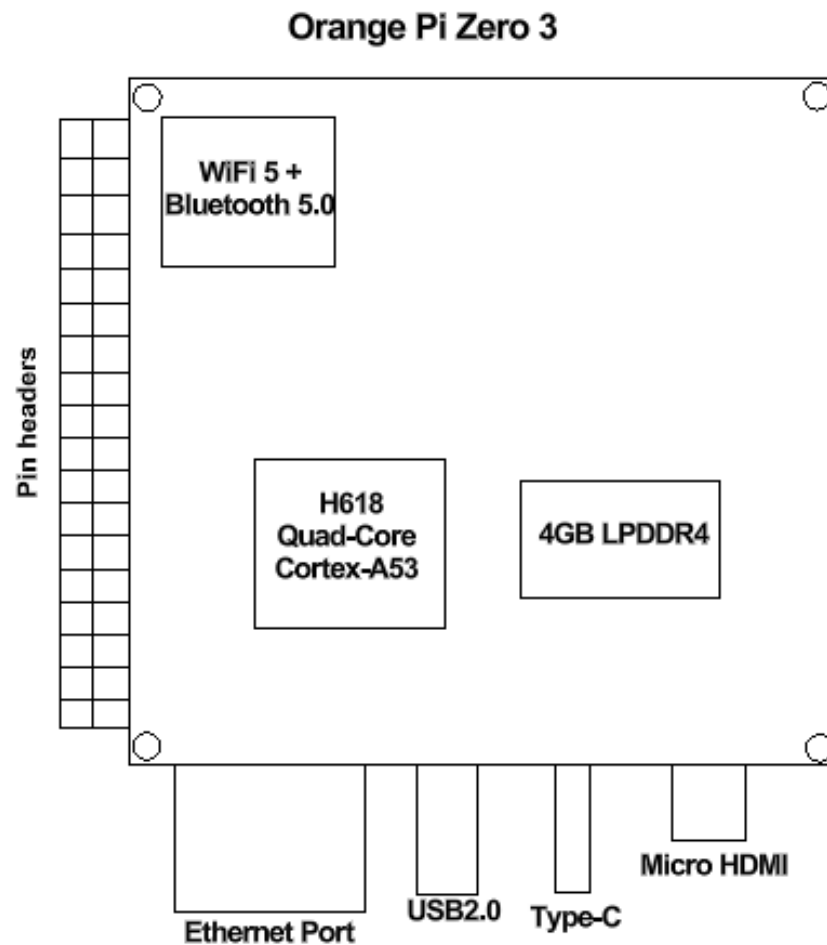


Fig 2. Orange Pi Zero

Figure 3 illustrates the DS3231 Real-Time Clock module, which serves as the authoritative and offline-capable timekeeping source for the attendance node. The module houses a highly accurate, temperature-compensated real-time clock integrated circuit that maintains accurate date and time information. The installed CR2032 coin cell battery provides backup power, ensuring that the timekeeping function continues accurately even during power interruptions, making the attendance logging fully resilient to power outages. It communicates with the microcontroller via the I2C pin header, utilizing the SDA and SCL connection points.

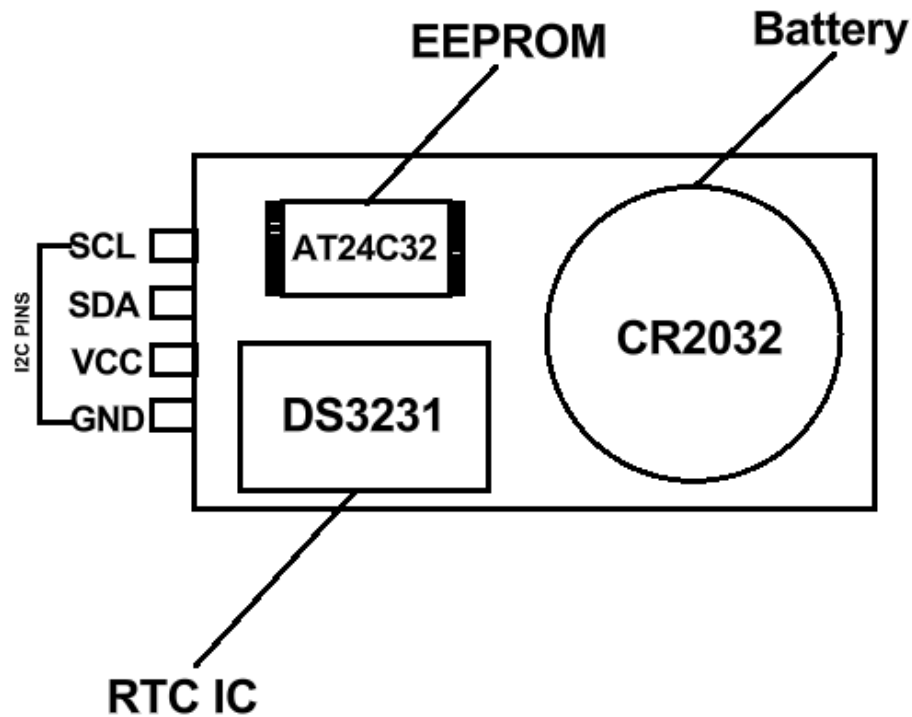


Fig. 3 DS3231 RTC MODULE

Figure 4 shows the I2C LCD 1602 Display module, a liquid-crystal display that provides local, real-time visual output of system status and attendance feedback directly at the scanning station. The 16x2 character screen displays alphanumeric information, such as prompts to place a finger on the scanner, successful authentication messages with the resident's name, or error notifications. The I2C backpack module soldered to the back of the LCD simplifies the wiring by requiring only four connections (VCC, GND, SDA, and SCL) to communicate with the microcontroller, significantly reducing GPIO pin usage and simplifying the hardware assembly.

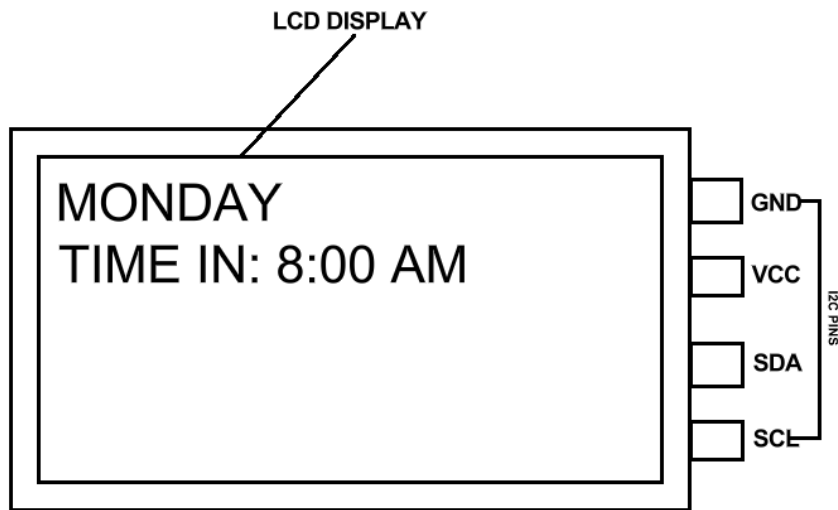


Fig 4. I2C LCD 1602 Display

Figure 5 depicts the ESP32S microcontroller, which functions as the primary processing brain for the biometric attendance terminal. The board features an integrated Wi-Fi module that enables wireless communication with the Orange Pi Zero local server over the local network, transmitting attendance logs and receiving biometric templates. Its numerous GPIO pins are utilized to interface with the fingerprint sensor, the LCD, and the RTC module. The powerful dual-core processor handles the complex tasks of orchestrating peripheral communication, managing network connections, and temporarily storing data in case of network disconnection.

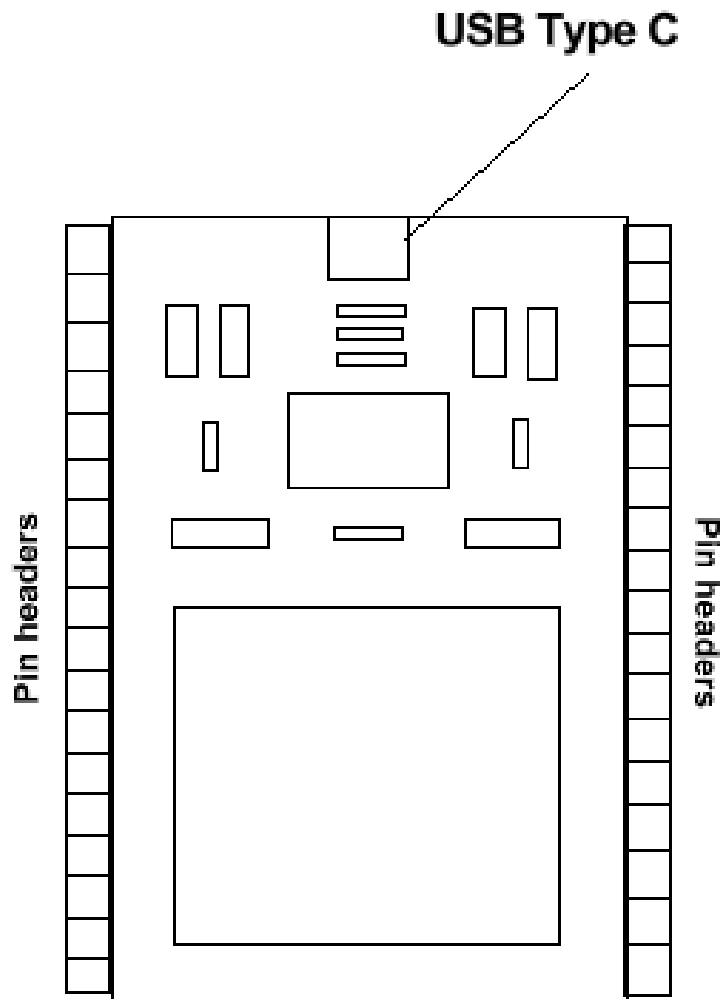


Fig 5. ESP32S

Figure 6 illustrates the R307 Fingerprint optical sensor, which serves as the primary biometric input device for the system. The scanning window utilizes an optical imaging mechanism to capture high-resolution images of a resident's fingerprint ridges and valleys. The built-in digital signal processing chip extracts minutiae points from the captured image and generates a unique biometric template. It communicates with the ESP32S microcontroller via a UART serial interface, allowing the system to enroll new fingerprints, search the local database for matches during attendance logging, and securely verify identities without relying on ID cards or passwords.

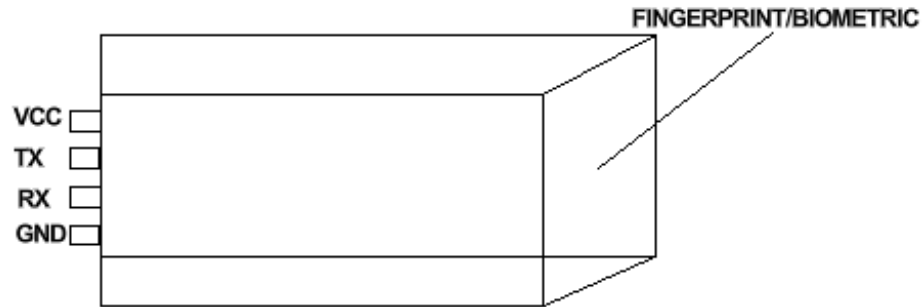


Fig 6. R307 Fingerprint

Figure 7 shows the web-based user interface dashboard of the system, accessible from any browser on the local area network by navigating to the server's IP address. The web dashboard is rendered using HTML, CSS, and JavaScript without requiring any app installation on the client devices. The main overview panel displays real-time statistics of resident populations and recent attendance activities. Navigation menus provide access to comprehensive modules for managing resident profiles, viewing detailed attendance records, issuing barangay certificates, and adjusting system configurations. The interface is designed to be intuitive and user-friendly, empowering barangay officials to efficiently oversee community operations from a centralized digital platform.

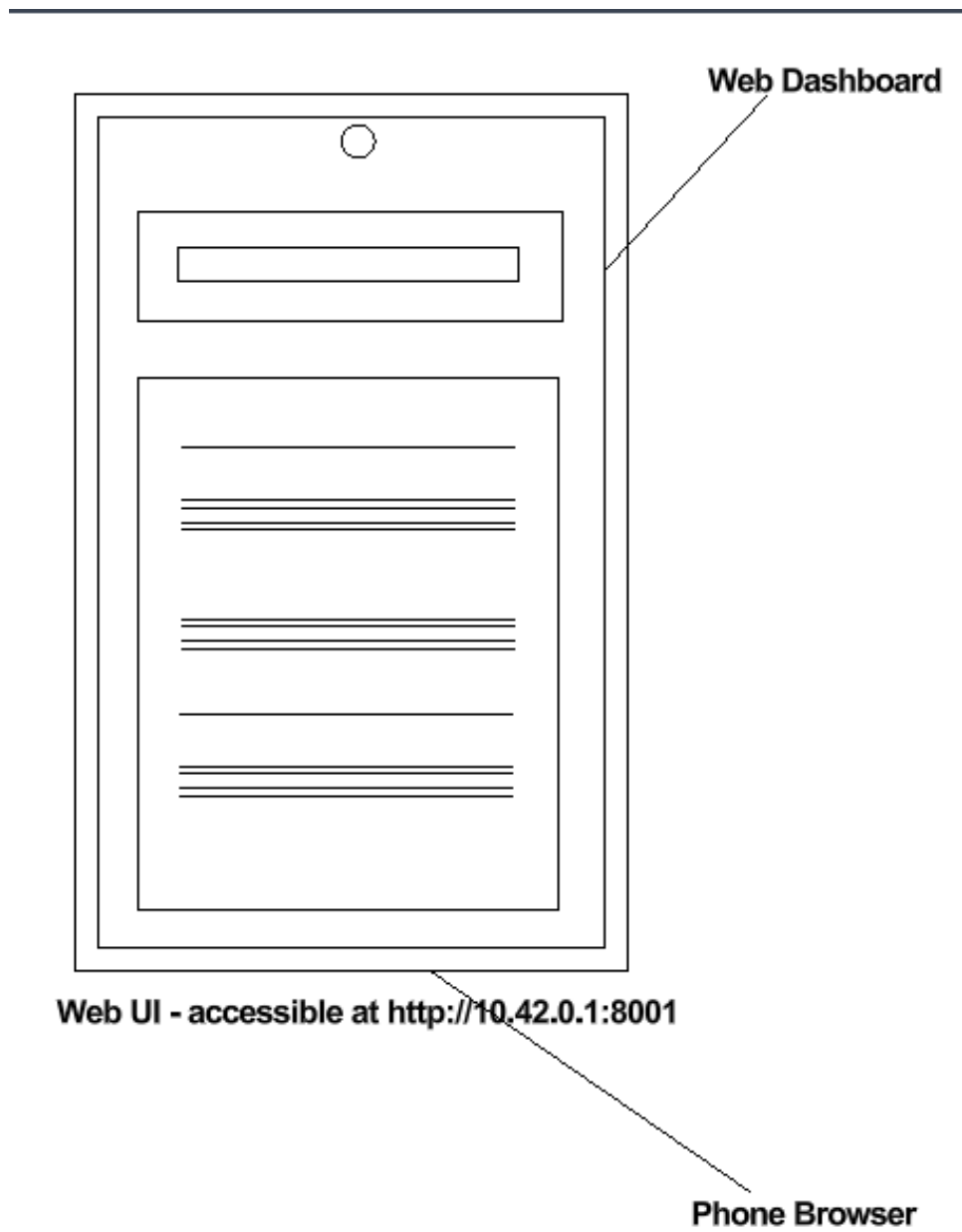


Fig 7. Web UI DASHBOARD